We work with the Fréchet log-likelihood:

$$\ell(A,\alpha) = T\log(A\alpha) - (1+\alpha)\sum_{t=1}^{T}\log(Ax_t) - \sum_{t=1}^{T}(Ax_t)^{-\alpha}.$$

✅ 1. First-order derivatives

Derivative w.r.t. A

Using
$\partial_A \log(A) = 1/A$,
$\partial_A (Ax)^{-\alpha} = -\alpha x (Ax)^{-\alpha-1}$,

$$\boxed{\frac{\partial \ell}{\partial A} = \frac{T}{A} - (1+\alpha)\frac{T}{A} + \alpha\sum_{t=1}^{T} x_t (Ax_t)^{-\alpha-1}}$$

Simplify:

$$\boxed{\frac{\partial \ell}{\partial A} = -\frac{\alpha T}{A} + \alpha\sum_{t=1}^{T} x_t (Ax_t)^{-\alpha-1}}$$

Derivative w.r.t. α

Using
$\partial_\alpha (Ax)^{-\alpha} = -(Ax)^{-\alpha}\log(Ax)$,

$$\boxed{\frac{\partial \ell}{\partial \alpha} = \frac{T}{\alpha} - \sum_{t=1}^{T}\log(Ax_t) + \sum_{t=1}^{T}(Ax_t)^{-\alpha}\log(Ax_t)}$$

✅ 2. Second-order derivatives

Second derivative w.r.t. A

$$\boxed{\frac{\partial^2 \ell}{\partial A^2} = \frac{\alpha T}{A^2} - \alpha(\alpha+1) \sum_{t=1}^{T} x_t^2 (A x_t)^{-\alpha-2}}$$

Second derivative w.r.t. α

$$\boxed{\frac{\partial^2 \ell}{\partial \alpha^2} = -\frac{T}{\alpha^2} - \sum_{t=1}^{T} (A x_t)^{-\alpha} \left[\log(A x_t)\right]^2}$$

Cross derivative

$$\boxed{\frac{\partial^2 \ell}{\partial A \partial \alpha} = -\frac{T}{A} + \sum_{t=1}^{T} x_t (A x_t)^{-\alpha-1} \left[1 - \alpha \log(A x_t)\right]}$$